

1 Solution — GIAM 7.4

Problem 7

1.1 Problem

Prove the binomial theorem:

$$\forall n \in \mathbb{N}, \quad \forall x, y \in \mathbb{R}, \quad (x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

(Convention: $\mathbb{N} = \{0, 1, 2, \dots\}$, and $0^0 = 1$ where it appears. Both interpretations are standard for this theorem.)

1.2 Proof I — Combinatorial

Write $(x + y)^n$ as the product of n identical factors

$$(x + y)^n = \underbrace{(x + y)(x + y) \cdots (x + y)}_{n \text{ factors}}.$$

Expand by *distributivity*: a term in the expansion is obtained by picking exactly one of the two summands (x or y) from each factor, then multiplying the picks together. Each pick is a sequence $(t_1, t_2, \dots, t_n)$ with each $t_i \in \{x, y\}$. There are 2^n such sequences, and the expanded sum is

$$(x + y)^n = \sum_{(t_1, \dots, t_n) \in \{x, y\}^n} t_1 t_2 \cdots t_n.$$

Each product $t_1 \cdots t_n$ collapses to a monomial $x^{n-k} y^k$, where k is the number of factors that picked y . Group the 2^n terms by their value of k :

$$(x + y)^n = \sum_{k=0}^n \left(\#\{\text{sequences with exactly } k \text{ y's}\} \right) \cdot x^{n-k} y^k.$$

The number of binary sequences of length n with exactly k y's is the number of ways to choose *which* k of the n positions hold y — a k -element subset of $\{1, \dots, n\}$. That is, $\binom{n}{k}$. Substituting,

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k. \quad \blacksquare$$

1.3 Demonstration — Combinatorial Proof for $n = 3$

For $n = 3$, there are $2^3 = 8$ sequences, grouped below by their number of y 's:

k	Sequences (each is a product $t_1 t_2 t_3$)	Count	Monomial
0	xxx	1	x^3
1	xyx, xyx, yxx	3	x^2y
2	xyy, yxy, yyx	3	xy^2
3	yyy	1	y^3

Table 1.1

Summing: $(x + y)^3 = 1 \cdot x^3 + 3 \cdot x^2y + 3 \cdot xy^2 + 1 \cdot y^3$, with coefficients $1, 3, 3, 1 = \binom{3}{0}, \binom{3}{1}, \binom{3}{2}, \binom{3}{3}$. ✓

1.4 Proof II — Induction on n

Induction on n , with Pascal's identity as the key step.

1.4.1 Base Case ($n = 0$)

$(x + y)^0 = 1$. The right-hand side is

$$\sum_{k=0}^0 \binom{0}{k} x^{0-k} y^k = \binom{0}{0} x^0 y^0 = 1 \cdot 1 \cdot 1 = 1.$$

Both sides equal 1. ✓

1.4.2 Inductive Step

Suppose the binomial theorem holds for some $n \geq 0$:

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k. \quad (\text{IH})$$

We show it also holds for $n + 1$. Starting from $(x + y)^{n+1} = (x + y) \cdot (x + y)^n$, apply (IH):

$$(x + y)^{n+1} = (x + y) \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

Distribute $(x + y)$ across the sum:

$$(x + y)^{n+1} = \sum_{k=0}^n \binom{n}{k} x^{(n+1)-k} y^k + \sum_{k=0}^n \binom{n}{k} x^{n-k} y^{k+1}.$$

In the second sum, *reindex* via $j = k + 1$ (so $k = j - 1$ and j runs from 1 to $n + 1$):

$$\sum_{k=0}^n \binom{n}{k} x^{n-k} y^{k+1} = \sum_{j=1}^{n+1} \binom{n}{j-1} x^{(n+1)-j} y^j.$$

Rename $j \rightarrow k$ to align indices, then combine the two sums by *splitting off* the boundary terms ($k = 0$ from the first, $k = n + 1$ from the second):

$$(x + y)^{n+1} = \underbrace{\binom{n}{0} x^{n+1}}_{=1} + \sum_{k=1}^n \left[\binom{n}{k} + \binom{n}{k-1} \right] x^{(n+1)-k} y^k + \underbrace{\binom{n}{n} y^{n+1}}_{=1}.$$

Apply **Pascal's identity** $\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$ (proven below) to the middle terms, and use $\binom{n}{0} = \binom{n+1}{0} = 1$ and $\binom{n}{n} = \binom{n+1}{n+1} = 1$ on the boundary terms:

$$(x + y)^{n+1} = \binom{n+1}{0} x^{n+1} + \sum_{k=1}^n \binom{n+1}{k} x^{(n+1)-k} y^k + \binom{n+1}{n+1} y^{n+1}.$$

This is exactly $\sum_{k=0}^{n+1} \binom{n+1}{k} x^{(n+1)-k} y^k$ — the binomial theorem for $n + 1$.

By induction, the binomial theorem holds for all $n \in \mathbb{N}$. ■

1.5 Lemma — Pascal's Identity

For $n \geq 1$ and $1 \leq k \leq n$,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Combinatorial proof. Both sides count k -subsets of $\{1, 2, \dots, n\}$. Partition the k -subsets by whether they *contain* the element n :

- **Contain n :** n is locked in; the remaining $k-1$ elements form a $(k-1)$ -subset of $\{1, \dots, n-1\}$ — there are $\binom{n-1}{k-1}$ such.
- **Do not contain n :** all k elements are chosen from $\{1, \dots, n-1\}$ — there are $\binom{n-1}{k}$ such.

Adding the two cases gives Pascal's identity. $\square$

Algebraic check. Direct factorial computation:

$$\binom{n-1}{k-1} + \binom{n-1}{k} = \frac{(n-1)!}{(k-1)!(n-k)!} + \frac{(n-1)!}{k!(n-1-k)!}.$$

Bring to a common denominator $k!(n-k)!$ and combine:

$$= \frac{k \cdot (n-1)!}{k!(n-k)!} + \frac{(n-k) \cdot (n-1)!}{k!(n-k)!} = \frac{(k+n-k)(n-1)!}{k!(n-k)!} = \frac{n!}{k!(n-k)!} = \binom{n}{k}. \checkmark$$

1.6 Remark — Two Proofs, Two Flavors

The two proofs above illustrate complementary styles:

Style	Strength	Limitation
Combinatorial	Reveals <i>what the coefficients count</i> . One-line idea: pick y from k of the n factors.	Requires comfort with “double counting” arguments.
Inductive	Mechanical; works in any commutative ring (not just $\mathbb{R}$). Uses only Pascal's identity.	Hides the combinatorial meaning behind index manipulation.

Table 1.2

The combinatorial proof is widely considered the more *illuminating*; the inductive proof is the more *formal*. Both are correct and instructive.

1.7 Remark — Where the Convention $0^0 = 1$ Bites

The expansion as written includes terms like $\binom{n}{0}x^ny^0$. When $y = 0$, the convention $0^0 = 1$ makes the $k = 0$ term reduce to x^n , recovering $(x + 0)^n = x^n$.

Without this convention, one would need to *exclude* the $k = 0$ and $k = n$ terms when $y = 0$ or $x = 0$ respectively, or treat $y = 0$ as a special case. The convention is harmless because the unique extension of y^k to $y = 0$ that makes the binomial expansion *continuous* in y at $y = 0$ is the one assigning $0^0 := 1$.

1.8 Remark — Generalizations

Multinomial theorem. For m summands:

$$(x_1 + x_2 + \cdots + x_m)^n = \sum_{\substack{k_1 + \cdots + k_m = n \\ k_i \geq 0}} \binom{n}{k_1, k_2, \dots, k_m} x_1^{k_1} x_2^{k_2} \cdots x_m^{k_m},$$

with $\binom{n}{k_1, \dots, k_m} = \frac{n!}{k_1! k_2! \cdots k_m!}$ the multinomial coefficient. The combinatorial proof generalizes verbatim: pick one of $x_1, \dots, x_m$ from each of the n factors; the number of ways to assign k_i factors to x_i (with $\sum k_i = n$) is the multinomial coefficient.

The $m = 3$ case $((x + y + z)^n)$ gives Pascal's tetrahedron from Problem 4.

Newton's binomial series. For non-integer (or even complex) exponents α and $|y| < |x|$:

$$(x + y)^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^{\alpha-k} y^k, \quad \binom{\alpha}{k} := \frac{\alpha(\alpha-1)\cdots(\alpha-k+1)}{k!}.$$

This generalizes the *finite* binomial expansion of integer n to an *infinite* series for arbitrary α , and is the foundation of formal power series and Taylor expansions of $(1 + y)^\alpha$.

Non-commutative variables. When x and y do *not* commute (e.g., matrices), the binomial expansion fails. The simplest counter-example: $(x + y)^2 = x^2 + xy + yx +$

$y^2 \neq x^2 + 2xy + y^2$ in general. The binomial theorem rests crucially on $xy = yx$ to collect $x^{n-k}y^k$ from the $\binom{n}{k}$ sequences with k y's. The Heisenberg-style generalization (e.g., quantum binomial theorem) replaces $\binom{n}{k}$ with q -binomial coefficients.

1.9 Remark — Symmetry $x \leftrightarrow y$

Both sides of the binomial theorem are *symmetric* under swapping x and y :

$$(x + y)^n = (y + x)^n.$$

This forces the coefficients on $x^{n-k}y^k$ and on $x^k y^{n-k}$ to match — i.e., $\binom{n}{k} = \binom{n}{n-k}$, the symmetric identity of Pascal's triangle. The binomial theorem doesn't *prove* this symmetry on its own (it's already built into the binomial coefficients), but it's a useful consistency check that the expansion respects the swap.